

Project Design Phase Solution Architecture

Date	11 July 2026
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process - with many sub-processes - that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram: Credit Card Approval Prediction System

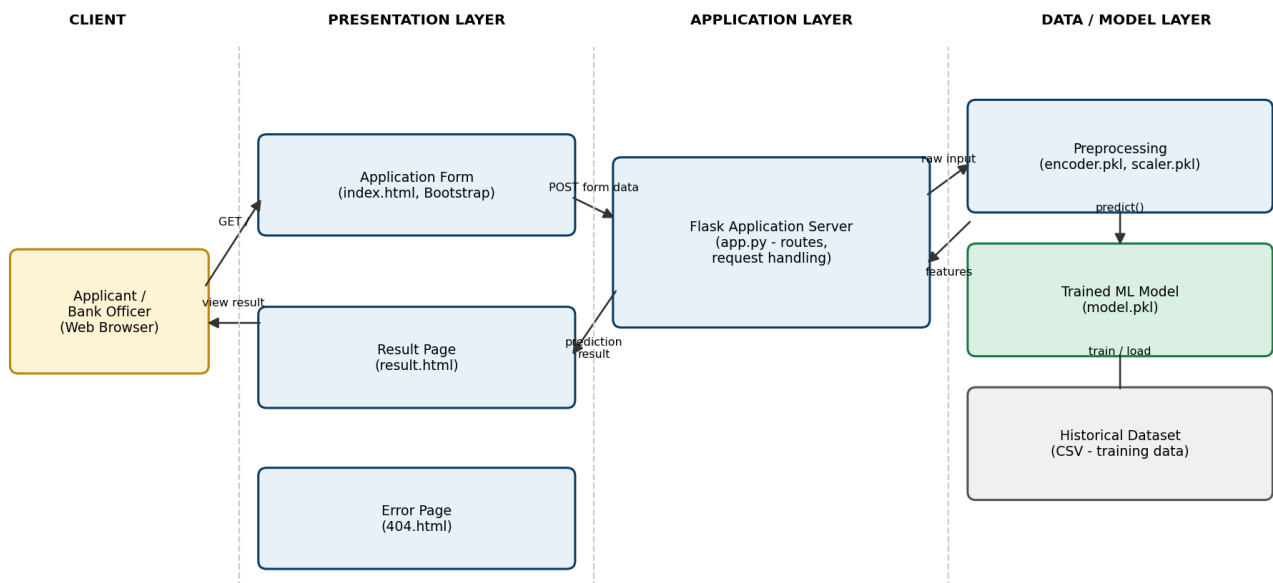


Figure 1: Architecture and data flow of the Credit Card Approval Prediction System, from applicant request through the Flask application, preprocessing pipeline, and trained ML model, back to the result page.

Layer-wise Description:

Layer	Description
Client	The applicant or bank officer interacts with the system through a standard web browser - no separate app installation required.
Presentation Layer	HTML/Bootstrap templates rendered by Flask: the application form, the prediction result page, and a custom error page for invalid routes.
Application Layer	The Flask server (app.py) handles incoming form submissions, routes requests, and coordinates between the presentation layer and the model layer.

Layer	Description
Data / Model Layer	Preprocessing objects (encoder, scaler) transform raw form input into model-ready features; the trained model (model.pkl) generates the Approved/Rejected prediction; the historical dataset is used offline during training and evaluation.